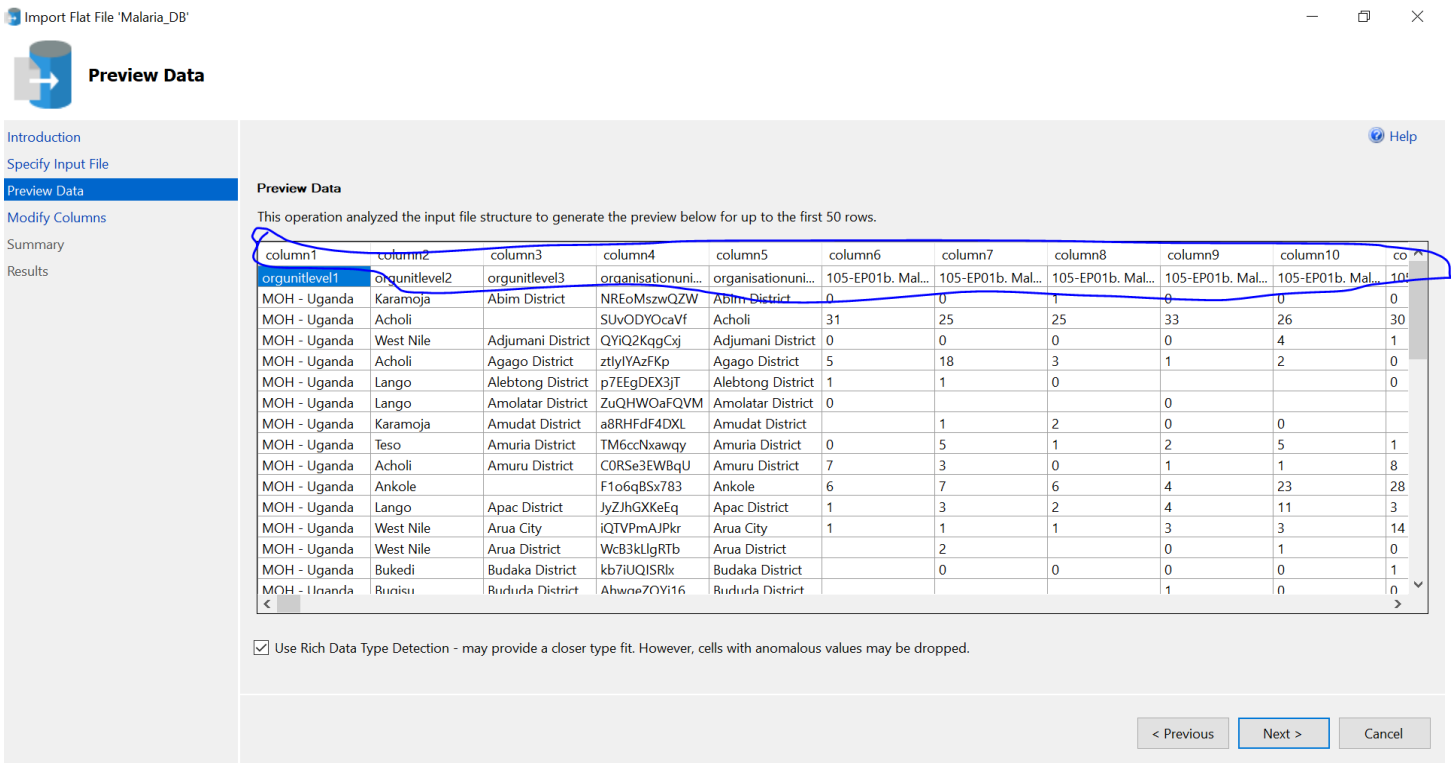
SQL SERVER, FILE IMPORTATION PROBLEM

PROBLEM 1 ⚠

- On Importation of the flat files (Csv files), SQL server normally misplaces our original column names to **row 2** with its own naming of columns, which makes it unsuitable for importation of large datasets that might contain cryptic/Variable column names.



column1	column2	column3	column4	column5	column6	column7	column8	column9	column10	co
orgunitlevel1	orgunitlevel2	orgunitlevel3	organisationuni...	organisationuni...	105-EP01b. Mal...	105-EP01b. Mal...	105-EP01b. Mal...	105-EP01b. Mal...	105-EP01b. Mal...	10
MOH - Uganda	Karamoja	Abim District	NREoMszwQZW	Abim District	0	0	1	0	0	0
MOH - Uganda	Acholi		SUvODYOcaVf	Acholi	31	25	25	33	26	30
MOH - Uganda	West Nile	Adjumani District	QYiQ2KqgCxj	Adjumani District	0	0	0	0	4	1
MOH - Uganda	Acholi	Agago District	ztlyIYAzFKp	Agago District	5	18	3	1	2	0
MOH - Uganda	Lango	Alebtong District	p7EEgDEX3jT	Alebtong District	1	1	0			0
MOH - Uganda	Lango	Amolatar District	ZuQHWOaFQVM	Amolatar District	0			0		
MOH - Uganda	Karamoja	Amudat District	a8RHFdF4DXL	Amudat District		1	2	0	0	
MOH - Uganda	Teso	Amuria District	TM6ccNxawqy	Amuria District	0	5	1	2	5	1
MOH - Uganda	Acholi	Amuru District	C0RSe3EWBqU	Amuru District	7	3	0	1	1	8
MOH - Uganda	Ankole		F1o6qBSx783	Ankole	6	7	6	4	23	28
MOH - Uganda	Lango	Apac District	JyZJhGXKeEq	Apac District	1	3	2	4	11	3
MOH - Uganda	West Nile	Arua City	iQTVPmAJPkr	Arua City	1	1	1	3	3	14
MOH - Uganda	West Nile	Arua District	WcB3kLlgRTb	Arua District		2		0	1	0
MOH - Uganda	Bukedi	Budaka District	kb7iUQISRlx	Budaka District		0	0	0	0	1
MOH - Uganda	Bugisu	Bududa District	AhwqeZOYi16	Bududa District				1	0	0

MANUAL ALTERATION OF COLUMN NAMES

⚠ **PROBLEM 2:**

- After importation of Raw Files in the Landing Zone(Malaria DB), We have to manually rename back the column names to the original column names which time consuming and Hectic. This method is not scalable for multiple large datasets.

```sql
--Renaming Columns

EXEC sp_rename 'malaria_data.[column1]',  'Region', 'COLUMN';
EXEC sp_rename 'malaria_data.[column2]',  'District', 'COLUMN';

EXEC sp_rename 'malaria_data1.[0-28_DAYS_Female]',  '0_28Days_Female',  'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_MALE]',  '0_28Days_Male',  'COLUMN';
EXEC sp_rename 'malaria_data1.[_0_28_DAYS_TOTAL]',  '0_28Days_Total',  'COLUMN';

EXEC sp_rename 'malaria_data1.[_10_19_YEARS_FEMALE]',  '10_19Days_Female', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19_YEARS_MALE]',  '10_19Days_Male',  'COLUMN';
EXEC sp_rename 'malaria_data1.[_10_19YRS_TOTAL]',  '10_19Days_Total',  'COLUMN';

EXEC sp_rename 'malaria_data1.[_20_YRS_FEMALE]',  '20+YRS_Female',  'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_YRS_MALE]', '20+YRS_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_20_YRS_TOTAL]', '20+YRS_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_5_9YRS_FEMALE]', '5_9YRS_Female',  'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_MALE]', '5_9YRS_Male', 'COLUMN';
EXEC sp_rename 'malaria_data1.[_5_9YRS_TOTAL]', '5_9YRS_Total', 'COLUMN';

EXEC sp_rename 'malaria_data1.[_29_DAYS_4YRS_FEMALE]',  '29Days_4Yrs_Female', 'COLUMN';
```

MANUAL CHANGE IN DATATYPES

⚠️ **PROBLEM3**

- Manual altering of columns to suitable datatypes for useability like **INT**, **float.** which is very hectic and non-scalable for big datasets for example a raw data set with more than 80 columns

```
...TICS\FULL PROJECT\SQL SCRIPTS\DataQuality Checks2.sql                    5
ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs(BSRDT)_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS(BSRDT)_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS(BSRDT)_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS(BSRDT)_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "TotalCasesConfirmed_BSRDT" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "0_28Days_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "10_19YRS_CT_Female" INT;
ALTER  TABLE malaria_data1 ALTER COLUMN "10_19YRS_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "10_19YRS_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "20+YRS_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_CT_Male" INT;
ALTER  TABLE malaria_data1 ALTER COLUMN "29Days_4Yrs_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS_CT_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS_CT_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "5_9YRS_CT_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Preg_10_19" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Preg_20Plus" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Preg_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "TotalCases_Female" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "TotalCases_Male" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "GrandTotalCases" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Under5_Female_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Under5_Male_Total" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Under5_GrandTotal" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "Area_km2" DECIMAL(18,8);
ALTER TABLE malaria_data1 ALTER COLUMN TotalPopulation DECIMAL(18,8);
ALTER TABLE malaria_data1 ALTER COLUMN "PopulationDensityPerKm2" DECIMAL
  (18,8);
ALTER TABLE malaria_data1 ALTER COLUMN "Latitude" DECIMAL(9,6);
ALTER TABLE malaria_data1 ALTER COLUMN "Longitude" DECIMAL(9,6);
ALTER TABLE malaria_data1 ALTER COLUMN "Year" INT;
ALTER TABLE malaria_data1 ALTER COLUMN "IncidenceRate" DECIMAL(18,8);
ALTER TABLE malaria_data1 ALTER COLUMN "PopulationDensity" DECIMAL
  (18,8);
```